EXAM SHIELD

Internet of Things and Embedded Systems Question Bank

Sensors, microcontrollers, IoT protocols and edge computing.

13 questions: 8 multiple choice, 3 fill in the gap and 2 written. Total 22 marks.
Edition: with answers (staff copy, ready to import into Exam Shield).

How this paper is laid out

Questions are numbered in one sequence. Multiple-choice options are lettered with a full stop; each multiple-choice question is worth one mark unless stated.

Fill-in-the-gap questions show each blank as a line; they earn one mark per blank, shown in square brackets, and are marked automatically.

Written questions show their marks in square brackets and are marked by an examiner.

To use this paper in Exam Shield, open an exam, choose Questions and then Import from PDF, and upload this file. Review the preview and import.

Section 1: Internet of Things and Embedded Systems

Sensors, microcontrollers, IoT protocols and edge computing.

1. Which lightweight publish/subscribe protocol is widely used in IoT?

A. SNMP
B. FTP
C. SMTP
D. MQTT

Answer: D

2. In MQTT, which component receives messages from publishers and passes them to subscribers?

A. Client
B. Gateway
C. Router
D. Broker

Answer: D

3. Which wireless technology is designed for low-power IoT links over several kilometres?

A. Bluetooth Classic
B. Wi-Fi 6
C. NFC
D. LoRaWAN

Answer: D

4. A microcontroller turns an analog sensor voltage into a number using a(n):

A. UART
B. DAC
C. ADC
D. PWM output

Answer: C

5. Which technique varies the duty cycle of a signal to control motor speed or LED brightness?

A. I2C
B. SPI
C. ADC
D. PWM

Answer: D

6. Which two-wire serial bus uses SDA and SCL lines to connect several devices?

A. I2C
B. UART
C. SPI
D. Ethernet

Answer: A

7. An embedded designer chooses a real-time operating system (RTOS) mainly for:

A. A graphical desktop
B. Built-in cloud storage
C. Deterministic task scheduling and timing
D. Unlimited memory

Answer: C

8. Why is CoAP suited to constrained IoT devices?

A. It only uses XML messages
B. It needs a persistent TCP connection
C. It runs over UDP with a compact binary header
D. It requires a web browser

Answer: C

9. Processing data close to where it is produced, instead of in a distant data centre, is called _____ computing. [1 mark]

Answer: edge

10. MQTT messages are published to named _____, and the highest MQTT QoS level is _____. [2 marks]

Answer: topics / topic | 2 / two

11. A _____ timer automatically resets a microcontroller if its software stops responding. [1 mark]

Answer: watchdog

12. Describe a layered IoT architecture (devices, gateway, cloud, application) for a smart-farming system, and explain where data should be processed and why. [5 marks]

13. Discuss three security challenges specific to IoT devices and one mitigation for each. [5 marks]